I. Introduction & Research Questions

Preterm birth, or delivery before 37 weeks of gestation, is a leading cause of death in children under five worldwide [1]. Current risk tools such as cervical length measurement and fetal fibronectin testing have variable accuracy [2], and they depend on equipment and medical expertise that may be scarce in low-resource settings. This matters because infant-survival outcomes depend heavily on *where* a baby is born. More than 90% of extremely preterm babies (<28 weeks) in low-income countries die within the first few days of life, versus less than 10% in high-income settings [1]. Predictive models built on cheap, noninvasive inputs could help identify high-risk pregnancies earlier. Electrohysterography (EHG) is one candidate. It records the uterus's electrical activity through abdominal skin electrodes, much like an ECG for heart activity [3]. Using an EHG signal, obstetricians can calculate the frequency of uterine contractions, length of each contraction, and contraction power of the uterus. These features indicate the condition of the uterus and are therefore potential markers of preterm risk [4].

The Term-Preterm EHG Database (TPEHG DB) [3] spawned many studies reporting near-perfect accuracy, but Vandewiele et al. [5] showed that these results were overly optimistic. Vandewiele reconstructed eleven predictive modelling pipelines, and found a key flaw in their methodologies; the studies applied over-sampling methods like SMOTE and ADASYN before partitioning the data into training and testing sets. This flaw leads to data leakage, and the synthetic observations derived from the training-set data ended up in the test-set, artificially inflating model accuracies. When corrected, Vandewiele found that AUCs fell from 0.9-0.99 down to roughly 0.47-0.65. However, Vandewiele et. al did not study clinical covariates or recording timing, and used a single filter setting. Our study aims to extend their findings with three research questions. Firstly, when oversampling, scaling, and feature selection are confined to training folds, how much predictive signal for preterm delivery remains in EHG features, with uncertainty quantified by repeated stratified cross-validation and bootstrap confidence intervals? Is it random chance, or do some trends exist? Secondly, how much are preterm outcomes explained by routine clinical variables (e.g., maternal age, diabetes, smoking status) and the gestational age at recording rather than by uterine activity itself? Finally, how sensitive are the results to preprocessing choices, including filter band, electrode channel, and selected features?

Our study will clarify whether the EHG signal has potential as a cheap, noninvasive predictor of preterm birth, and how much EHG-based predictions are influenced by clinical measurements and data preprocessing.

II. Data Sources

For this study, we will use the open-access TPEHG DB v1.0.1 on PhysioNet. The dataset contains 1300 recordings from 300 pregnancies, one per woman, collected at University Medical Centre Ljubljana from 1997 to 2005. The collection includes recordings from 262 women who had full-term pregnancies and 38 whose pregnancies ended prematurely with 162 of the recordings made before the 26th week of gestation, and 138 later. Importantly, the dataset has collected clinical parameters (ie. pregnancy duration, gestation duration, maternal age, hypertension, diabetes, etc.), EHG signal data, and labelled outcomes for preterm and full-term birth. This dataset is highly relevant to our research questions as it is the benchmark the previously mentioned studies used, so we can replicate and correct their pipelines on identical data. The labelled outcomes allow supervised modelling, the signal data supports testing different preprocessing choices, and the clinical variables let us measure how much of the predictive performance comes from electrical uterine activity versus clinical parameters.

Outcome: 262 term, 38 preterm (12.7% preterm)
Recording timing: 162 before week 26, 138 at or after; preterm split evenly 19 / 19
Signals: 3 bipolar channels from 4 abdominal electrodes, 30 min each
Sampling: 20 Hz, 16-bit
Pre-filtered versions: Butterworth band-pass at 0.08-4 Hz, 0.3-3 Hz, 0.3-4 Hz; drop first and last 180 s of filtered signals
Clinical variables: Maternal age, parity, prior abortions, weight, hypertension, diabetes, placental position, bleeding, funneling, smoking
Precomputed features: RMS, median frequency, peak frequency, sample entropy per filtered channel
Exclusions by curators: Records with artifacts, induced labor, Cesarean delivery

III. Proposed Methods

Course topic	How we use it	Key concepts	Project stage
Data Sampling and Bias & Variance (Unit 6)	Compare oversampling before vs inside training folds and quantify how leakage inflates performance	SMOTE and random oversampling, class weights, bias-variance under small n	Data analysis (RQ1)
Normalization, Missing & Outlier, Feature Selection (Units 1, 7, 5)	Standardize features on training folds only, handle missing clinical fields and artifact segments, and select features inside folds	Fold-wise scaling, imputation, outlier handling, filter and embedded selection	Data cleaning
PCA & t-SNE (Units 4, 2)	Visualize term vs preterm separation and how features track recording week	Variance explained, 2D embeddings for visualization only	Data investigation
Logistic Regression, SVM, Random Forest (Units 2, 4, 6)	Baseline and comparison classifiers for preterm vs term	Regularized logistic regression (baseline), kernel SVM, random forest	Data analysis
Splits & Cross-Validation, ROC-AUC, Statistical Evaluation, Uncertainty & Calibration, Feature Importance (Units 2-6)	Leak-free estimates with uncertainty and which features & variables carry signal (RQ2, RQ3)	Repeated stratified CV, ROC and PR curves, bootstrap CIs, permutation test, calibration plots, permutation feature importance	Analysis, final report

IV. References

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